

3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Experiment 3.1

```
clc;
```

```
clear;
```

```
clf;
```

```
//-----
```

```
// Addition and Multiplication of Two Sequences
```

```
//-----
```

```
N = 30;          // Number of samples
```

```
n = 0:N-1;       // Discrete time index
```

```
// Sequence 1: Sinusoidal Sequence
```

```
f1 = 0.05;       // Frequency (cycles/sample)
```

```
A1 = 2;          // Amplitude
```

```
x1_n = A1 * sin(2 * %pi * f1 * n);
```

```
// Sequence 2: Exponentially Decaying Sequence
```

```
a2 = 0.1;
```

```
A2 = 3;
```

```
x2_n = A2 * exp(-a2 * n);
```

```
// Addition
```

```
x_add = x1_n + x2_n;
```

```
// Multiplication
```

```
x_mult = x1_n .* x2_n;
```

```
//-----
```

```
// Plot the Signals
```

```
//-----
```

```
subplot(4,1,1);
```

```
plot2d3(n, x1_n);
```

```
xlabel("n");
```

```
ylabel("Amplitude");
```

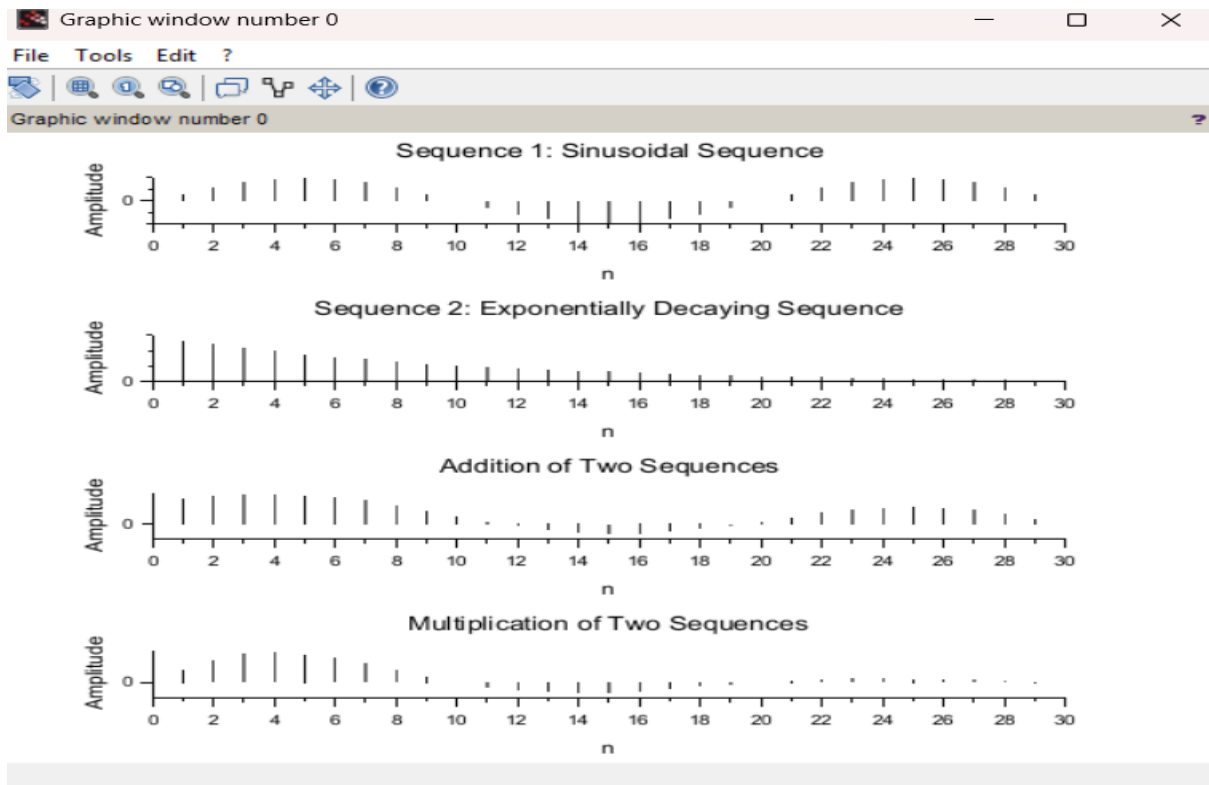
```
title("Sequence 1: Sinusoidal Sequence");
```

```
subplot(4,1,2);
```

```
plot2d3(n, x2_n);  
xlabel("n");  
ylabel("Amplitude");  
title("Sequence 2: Exponentially Decaying Sequence");
```

```
subplot(4,1,3);  
plot2d3(n, x_add);  
xlabel("n");  
ylabel("Amplitude");  
title("Addition of Two Sequences");
```

```
subplot(4,1,4);  
plot2d3(n, x_mult);  
xlabel("n");  
ylabel("Amplitude");  
title("Multiplication of Two Sequences");
```



Experiment 3.2

clc;

clear;

// Time vector

t = -10:0.001:10;

dt = t(2) - t(1);

//-----

// Energy Signal

// $x_1(t) = e^{-2|t|}$

//-----

```
x1 = exp(-2 * abs(t));
```

```
E1 = sum(abs(x1).^2) * dt;
```

```
P1 = E1 / (max(t) - min(t));
```

```
//-----
```

```
// Power Signal
```

```
// x2(t) = cos(2π5t)
```

```
//-----
```

```
x2 = cos(2 * %pi * 5 * t);
```

```
E2 = sum(abs(x2).^2) * dt;
```

```
P2 = E2 / (max(t) - min(t));
```

```
//-----
```

```
// Display Results
```

```
//-----
```

```
disp("Energy of x1 = " + string(E1));
```

```
disp("Average Power of x1 = " + string(P1));
```

```
disp("Energy of x2 = " + string(E2));
```

```
disp("Average Power of x2 = " + string(P2));
```

```
"Energy of x1 = 0.5000007"  
"Average Power of x1 = 0.025"  
"Energy of x2 = 10.001"  
"Average Power of x2 = 0.50005"  
  
-->
```

Experiment 3.3

// Test Case: Autocorrelation of a Rectangular Pulse Sequence

// Define parameters

N = 20; // Number of samples

x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5

// Compute the autocorrelation

R_xx = correl(x_n, x_n);

// Create a lag vector for plotting

lags = -N+1:N-1;

// Plot the input sequence

clf; // Clear the current figure

subplot(2,1,1);

bar(0:N-1, x_n, 'b');

xlabel('n (Discrete Time Index)');

ylabel('Amplitude');

title('Input Sequence: Rectangular Pulse Sequence');

```
// Plot the autocorrelation sequence
```

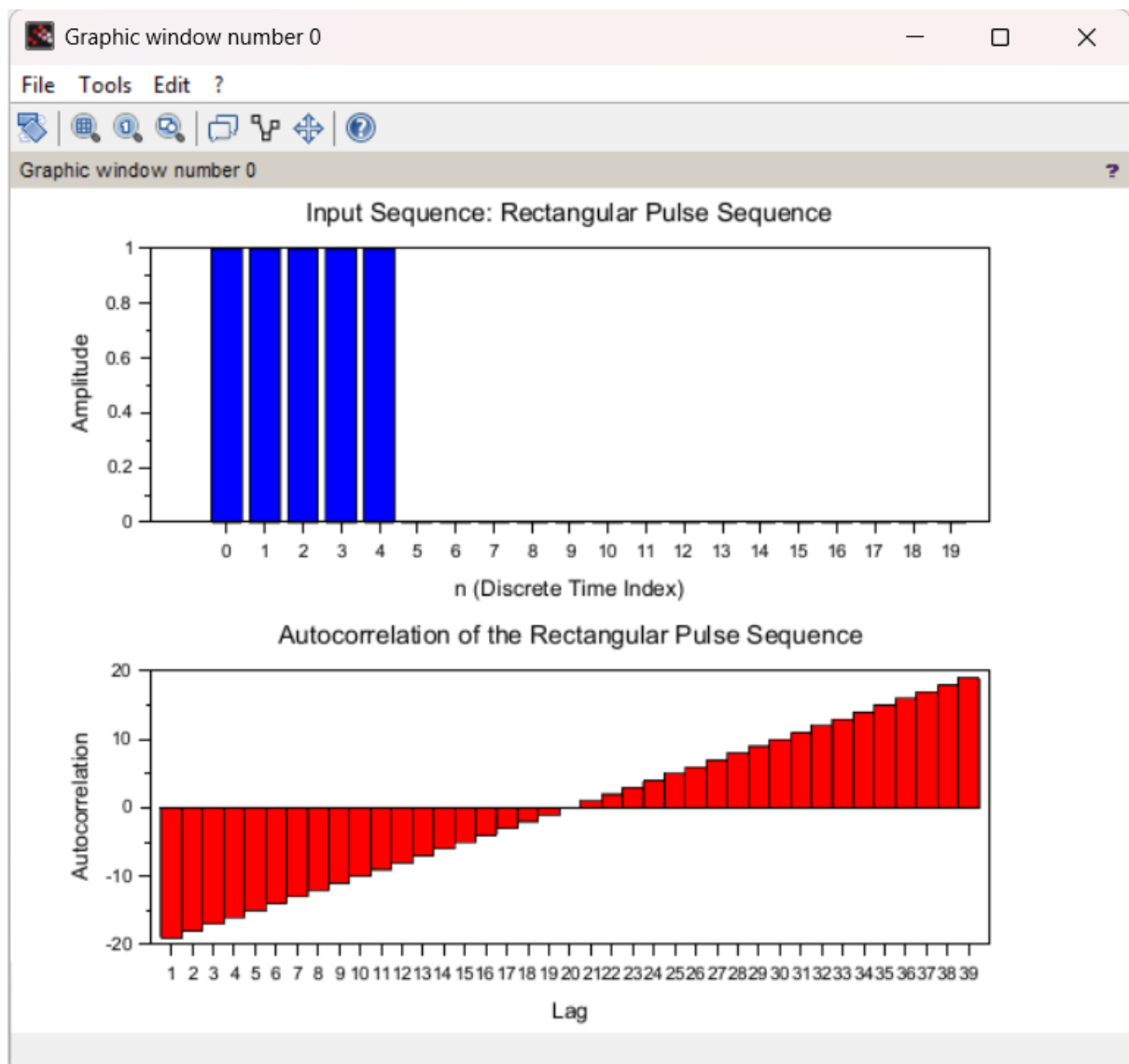
```
subplot(2,1,2);
```

```
bar(lags, R_xx, 'r');
```

```
xlabel('Lag');
```

```
ylabel('Autocorrelation');
```

```
title('Autocorrelation of the Rectangular Pulse Sequence');
```



Experiment 3.4

// Test Case: Cross-Correlation of Two Discrete-Time Sequences

// Define parameters

N = 20; // Number of samples

x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5

delay = 3; // Delay of 3 samples

y_n = [zeros(1,delay), x_n(1:\$-delay)]; // Delayed version of x[n]

// Compute the cross-correlation

R_xy = correl(x_n, y_n);

// Create a lag vector for plotting

lags = -N+1:N-1;

// Plot the first sequence

clf; // Clear the current figure

subplot(3,1,1);

bar(0:N-1, x_n, 'b');

xlabel('n (Discrete Time Index)');

ylabel('Amplitude');

title('Sequence 1: Rectangular Pulse Sequence');

// Plot the second sequence


```

subplot(3,1,2);
bar(0:N-1, y_n, 'g');
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Sequence 2: Delayed Rectangular Pulse Sequence');
// Plot the cross-correlation sequence
subplot(3,1,3);
bar(lags, R_xy, 'r');
xlabel('Lag');
ylabel('Cross-Correlation');
title('Cross-Correlation between Sequence 1 and Sequence 2');

```

